NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_*CO-SC(MODEL-A& APEX) Date:30-12-24

Time: 3HRS GTM-6 Max. Marks: 300

KEYSHEET

PHYSICS

1	2	3	4	5	6	7	8	9	10
A	B	D	C	B	A	D	C	D	A
11	12	13	14	15	16	17	18	19	20
A	D	A	B	C	A	D	B	A	C
21	22	23	24	25					
8	3	3	8	9					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
C	D	D	B	C	C	A	A	B	C
36	37	38	39	40	41	42	43	44	45
B	C	C	A	D	B	D	C	D	D
46	47	48	49	50					
18	2	163	8	5					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
C	B	C	D	A	D	D	A	D	D
61	62	63	64	65	66	67	68	69	70
A	C	B	A	B	A	B	A	A	A
71	72	73	74	75					
45	149	16	20	25					

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HINTS&SOL

PHYSICS

1. $\left[\frac{\alpha Z}{k\theta}\right]=\left[M^0L^0T^0\right]\Rightarrow[\alpha]=\left[\frac{k\theta}{Z}\right]$

Further, $[P]=\left[\frac{\alpha}{\beta}\right]\Rightarrow[\beta]=\left[\frac{\alpha}{P}\right]=\left[\frac{k\theta}{ZP}\right]$

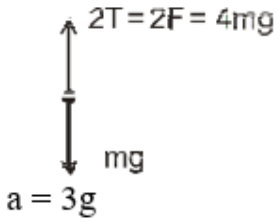
Dimensions of $k\theta$ are that of energy. Hence,

$[\beta]=\left[\frac{ML^2T^{-2}}{LML^{-1}T^{-2}}\right]=\left[M^0L^2T^0\right]$

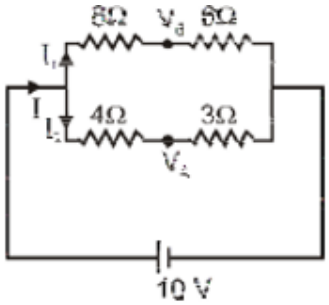
Therefore, the correct option is (A).

2. D. of block (FBD)

$\Rightarrow 2T-mg=mg$



3.



$R_{eq}=\frac{14\times7}{14+7}=\frac{14}{3}\Omega$

Net current $I=\frac{10}{\frac{14}{3}}=\frac{15}{7}A$

From Current division

$I_1=\frac{7}{21}\times\frac{15}{7}=\frac{5}{7}A$

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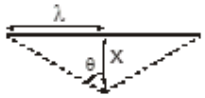
$I_2 = I - I_1 = \frac{15}{7} - \frac{5}{7} = \frac{10}{7}$

From KVL

$8I_1 + V_B - V_B - 4I_2 \rightarrow V_A - V_B = 8 \times - = 0$

4. New length

$\lambda' = 2\left(\sqrt{l^2 + x^2}\right)$



$\text{Strain} = \frac{\Delta l}{l} = \frac{l' - 2l}{2l} \Rightarrow \frac{2k\left(1 + \frac{x}{2l^2}\right) - 2l}{2l} = \frac{x^2}{2l^2}$

5. $248nm \equiv 1240 / 248ev = 5ev$

$310nm = 1240 / 310ev = 4ev$

$\frac{K.E_1}{K.E_2} - \frac{4}{1} - \frac{5ev - W}{4ev - W}$

$\Rightarrow 16 - 4W = S - W \Rightarrow 11 = 3W \Rightarrow W = \frac{11}{3}$

$= 3.67ev \cong 3.7ev$

6. Since there are six transition

${}^nC_2 = 6 \Rightarrow n = 4$

7. $W_G + W_f = 0 - 0$

$10 \times 1 + W_f = 0$

$10 - \mu mgx = 0$

$x = 5m$

8. $E_x = 0;$

$E_y = 2.5\frac{N}{C}\cos\left[\left(2\pi \times 10^6 \frac{rad}{s}\right)t - \left(\pi \times 10^{-2} \frac{rad}{m}\right)x\right]$

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$E_z = 0$

Comparing the given equation

$$E_y = 2.5\frac{N}{C}\cos\left[\left(2\pi\times10^6\,\frac{rad}{\sec}\right)t-\left(\pi\times10^{-2}\,\frac{rad}{\sec}\right)x\right]$$

With the standard equation

$E_y = E_0\cos(\omega t - kx)$

We get

$\omega = 2\pi f = 2\pi\times10^6$

$\therefore f = 10^6\,Hz$

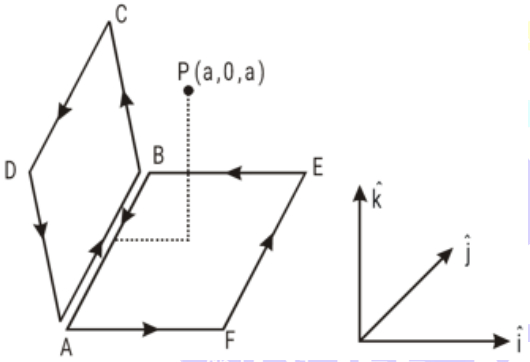
Moreover, we know that

$\frac{2\pi}{\lambda} = k = \pi\times10^{-2}\,m^{-1}$

$\Rightarrow \lambda = 200m$

9. The magnetic field at $P(a,0)$ due to the loop is equal to the vector sum of the magnetic field produced by loops $ABCDA$ and $AFEBA$ as shown in the figure.

Magnetic field due to loop $ABCDA$ will be along $\hat{i}$ and due to loop $AFEBA$, along $\hat{k}$. Magnitude of magnetic field due to both the loops will be equal. Therefore, direction of a resultant magnetic field at P will be $\frac{1}{\sqrt{2}}(\hat{i}+\hat{k})$



10. Position of 1st minima

$$Y = \frac{\lambda D}{a} = \frac{(5\times10^{-5})(0.6)}{0.02\times10^{-2}}$$

$y = 0.15cm$

11. $P_{out} = 100\ watt$

$P_{in} = 200\times0.6\,watt. = 120watt$

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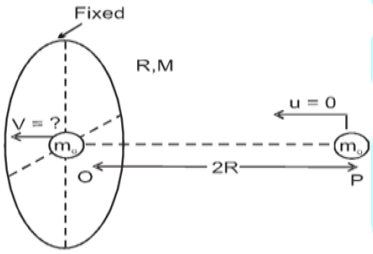
$$\eta\% = \frac{100}{120} \times 100 = \frac{500}{6} = 83.3\%$$

12. Applying energy conservation form P to O

$K_i + U_i = K_r + U_r$

$$O + (M_0)\left(\frac{-GM}{\sqrt{R^2 + (2R)^2}}\right) = \frac{1}{2} m_0 V^2 + (m_0)\left(\frac{-GM}{R}\right)$$

$$V = \sqrt{\frac{2GM}{R}\left(1 - \frac{1}{\sqrt{5}}\right)}$$



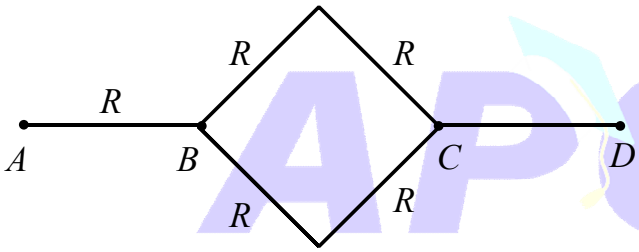
13. $R_t = R_0[1 + \alpha \Delta t]$

14. Diode in reverse by so current through A_1 is zero.

15. Let the thermal resistance of each rod be R

The two resistances connected along two paths from B to C are equivalent to $2R$ each and their parallel combination is R.

Effective thermal resistance between B and $D = 2R$

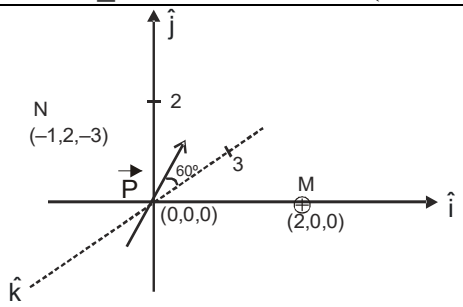


Temperature of interface $\theta = \frac{R_1\theta_1 + R_2\theta_1}{R_1 + R_2}$

$$\theta = \frac{R \times 20 + 2R \times 200}{R + 2R} = \frac{420}{3} = 140°C$$

16.

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$$V = K\frac{\vec{p}.\vec{r}}{r^3} = K.\frac{\left(\hat{i}+3\hat{j}\right).\left(-\hat{i}+\sqrt{3}\hat{j}+0\hat{k}\right)}{\left((-1)^2+\left(\sqrt{3}\right)^2+(0)^2\right)}$$

$$\therefore V = K.\frac{2}{8} = \frac{K}{4}$$

17. $F = \dfrac{2mV}{\Delta t}$

18. Angular momentum $L = (m \to v\cos 45°)h$

Where $h = v^2 \sin^2 452g$

$$|L| = \left(mv\sqrt{2}\right) \times v^2 \sin^2 452g = \frac{mv^3}{4\sqrt{2}g}$$

19. $P_{gas} = \dfrac{1}{3}\dfrac{mN}{V}V_{rms}^2$ …(i)

$$P_1 = \frac{1}{3}\frac{\left(\frac{m}{2}\right)N}{V}\left(2V_{rms}\right)^2 \quad \ldots(ii)$$

Dividing equation (i) and (ii) $P_1 = 2P_0$

20. With dc : $P = V^2/R \Rightarrow R = (100)/20 = 5\Omega$;

With ac: $P = V_{rms}^2 R/Z^2 \Rightarrow Z^2 = (10)^2 \times 5/10 = 50\Omega$

Also $Z^2 = R^2 + 4\pi^2\sqrt{2}L^2$

$\Rightarrow 50 = (5)^2 + 4(3.14)^2\sqrt{2}\left(10\times10^{-3}\right)^2 \Rightarrow v = 80Hz$

21. $F = BiL \quad L = 2r$

$= Bi(2r)$

$= 1\times2\times4\times10^{-2} = 8\times10^{-2}N$

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22. $I=\frac{8-4}{9+1+2}=\frac{1}{3}A$

$V_o-1\times I+4-8-2I-V_Q=0$

$V_P-V_Q=3V$

23. Energy of $E=h\nu=E_0z^2\left[\frac{1}{(n-1)^2}-\frac{1}{n^2}\right]$

$h\nu\approx E_0z^2\left[\frac{2n}{n^4}\right]\Rightarrow\nu\propto\frac{1}{n^3}$

24. $a=\frac{40}{12}=\frac{10}{3}m/s^2$

$T_1=\frac{40}{3}N$ $\quad$ $T_2=\frac{80}{3}N$

Strain in wire $2=\frac{80}{3\times5\times10^{-7}\times2\times10^{11}}=\frac{8}{3}\times10^{-4}$

25. $\frac{1}{f}=\frac{1}{V}+\frac{1}{4}=\frac{1}{-12}=\frac{1}{V}+\frac{1}{-20}$

$\Rightarrow\frac{1}{V}=\frac{1}{20}-\frac{1}{12}$

$\Rightarrow V=-30\ cm$

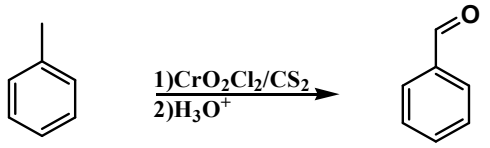
Velocity of image $\frac{dV}{dt}=-\left(\frac{V^2}{u^2}\right)\frac{du}{dt}=-\left(\frac{-30}{-20}\right)^2\times4=9cm/s$

$\Rightarrow 9\ cm/\text{sec}$ away from mirror.

CHEMISTRY

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27. 

This is called etard reaction , is used to convert toluene into benzaldehyde.

28.

It is non reducing sugar and it does not give silver mirror with tollens reagent.

29. Covalent radius / pm[a] | 70 | 110 | 121 | 141 | 148

30.

is a aromatic compound.

31. $Ac = 6d^1 7s^2 \text{ -------- } Ac^{+3} = 6d^0 7s^0$

$Lr = 5f^{14} 6d^1 7s^2 \text{ -------- } Lr^{+3} = 5f^0 6d^0 7s^0$

32. B = 2 ; Al = 1.5 ; Ga = 1.6 ; In = 1.7 ; Tl = 1.8

33. Since $n_i = 5$ and $n_f = 2$, this transition gives rise to a spectral line in the visible region of the

Balmer series. From equation (2.17)

$$\Delta E = 2.18 \times 10^{-18} J \left[\frac{1}{5^2} - \frac{1}{2^2} \right] = -4.58 \times 10^{-19} J$$

$$v = \frac{\Delta E}{h} = \frac{4.58 \times 10^{-19} J}{6.626 \times 10^{-34} Js} = 6.91 \times 10^{14} Hz$$

$$\lambda = \frac{c}{v} = \frac{3.0 \times 10^8 ms^{-1}}{6.91 \times 10^{14} Hz} = 434 nm$$

34.

Is a tertiary alcohols

35.

3-ehtyl-5-fluoro-2-formyl-2-hydroxy-4-methylhexa-3,5-dienoic acid

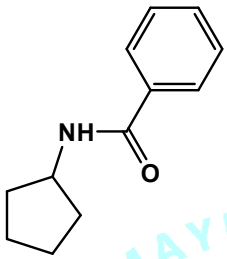
36. NH_3 = pyramidal

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IF_7 = pentagonal bipyramidal

$Ni(CO)_4$ = Tetrahedral

PCl_5 = Trigonal bipyramidal

37. **FINAL PRODUCT =** 

38. X = Zn – Hg + HCl Y=EtONa + EtOH

39. $CH_2 = CHCl$ is an example of vinyl halide and halogen atom is attached to sp^2 hybridized carbon atom

40. Solubility of gases in liquids decreases with rise in temperature

41. Ti = - 1.63
Cr = - 0.90
Cu = + 0.34
Zn = - 0.76

42.

43.

44. The sodium fusion extract is acidified with nitric acid and then treated with silver nitrate. A white precipitate, soluble in ammonium hydroxide shows the presence of chlorine

45. $[Ag(NH_3)_2][Ag(CN)_2]$ is named as: diamminesilver(I) dicyanidoargentate(I)
$[Ni(CO)_4]$ has tetrahedral geometry but is diamagnetic
$[Co(NH_3)_5(SO_4)]Br$ and $[Co(NH_3)_5Br]SO_4$ are ionization isomers
$[Co(en)_3]^{+3}$ has two optical isomers

46. $$\log\frac{k_2}{k_1} = \frac{E_a}{2.303R}\left[\frac{T_2 - T_1}{T_1T_2}\right]$$

$$\log\frac{0.07}{0.02} = \left(\frac{E_a}{2.303\times 8.314\,JK^{-1}mol^{-1}}\right)\left[\frac{700-500}{700\times 500}\right]$$

$$0.544 = E_a \times 5.714 \times 10^{-4}/19.15$$

$$E_a = 0.544 \times 19.15/5.714 \times 10^{-4} = 18230.8\ J$$

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47. Let the solubility of $Ni(OH)_2$ be equal to S.

Dissolution of S mol/L of $Ni(OH)_2$ provides S mol/L of Ni^{2+} and 2S mol/L of OH^-, but the total concentration of OH^- = (0.10 + 2S) mol/L because the solution already contains 0.10 mol/L of OH^- from NaOH.

$K_{sp} = 2.0 \times 10^{-15} = [Ni^{2+}]\,[OH^-]^2$

$= (S)\,(0.10 + 2S)^2$

As K_{sp} is small, $2S << 0.10$,

thus, $(0.10 + 2S) \approx 0.10$

Hence,

$2.0 \times 10^{-15} = S\,(0.10)^2$

$S = 2.0 \times 10^{-13}\ M = [Ni^{2+}]$

48. We know $\Delta_r G^\ominus = -2.303\ RT \log K_p$ and $R = 8.314\ JK^{-1}\ mol^{-1}$

Therefore, $\Delta_r G^\ominus =$

$-2.303\ (8.314\ J\ K^{-1}\ mol^{-1}) \times (298\ K)\ (\log 2.47 \times 10^{-29})$

$= 163000\ J\ mol^{-1}$

$= 163\ kJ\ mol^{-1}$.

49. $$\text{Retardation factor} = \frac{\text{Distance travelled by sample/organic compound}}{\text{Distance travelled by solvent}}$$

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50. $pV = nRT$

$p \times 1L = 0.15 \text{ mol} \times 0.083 \text{ bar L mol}^{-1}\text{K}^{-1} \times 400 \text{ K}$

$p = 4.98$ bar

$$N_2O_4 \rightleftharpoons 2NO_2$$

Initial pressure: 4.98 bar 0

At equilibrium: (4.98 – x) bar 2x bar

Hence,

p_{total} at equilibrium = $p_{N_2O_4} + p_{NO_2}$

$9.15 = (4.98 - x) + 2x$

$9.15 = 4.98 + x$

$x = 9.15 - 4.98 = 4.17$ bar

Partial pressures at equilibrium are,

$p_{N_2O_4} = 4.98 - 4.17 = 0.81$bar

$p_{NO_2} = 2x = 2 \times 4.17 = 8.34$ bar

$K_p = (p_{NO_2})^2 / p_{N_2O_4}$

$= (8.34)^2/0.81 = 85.87$

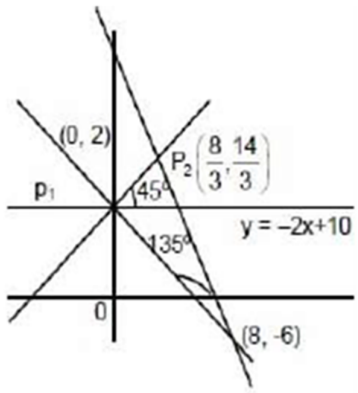
$K_p = K_c(RT)^{\Delta n}$

$85.87 = K_c(0.083 \times 400)^1$

$K_c = 2.586 = 2.6$

MATHEMATICS

51.



L_2 is $y = -x + 2$

$$\text{Area of triangle} = \frac{1}{2}\begin{vmatrix} 0 & 2 & 1 \\ \frac{8}{3} & \frac{14}{3} & 1 \\ 8 & -6 & 1 \end{vmatrix} = \frac{64}{3}$$

52. $(2023+2)^n = 7k + 2^n$

$n = 2024^{2023} = (205-1)^{2023} = 2q - 1 = 3p + 2$

$7k + 2^n = 7k + 4(1+7)^p = 7m + 4$

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53. $\because z.\bar{z} = |z|^2$

$z^{2022} = \bar{z}$

$\Rightarrow |z|^{2022} = |\bar{z}| = |z|$

$\Rightarrow |z|\left(|z|^{2021} - 1\right) = 0$

$\Rightarrow (0,0)$ and $|z|^{2021} = 1$

$\Rightarrow |z| = 1$

$\therefore z^{2022} = \bar{z}$

$\Rightarrow z^{2023} = z\bar{z} = 1$

$\therefore z^{2023} = 1$

$\Rightarrow$ 2023 distinct solutions

$\therefore$ So total (x, y) pairs are N = 2024

54. We have,

$$\vec{c} = x\vec{a} + y\vec{b} + \vec{a} \times \vec{b} \quad \text{and} \quad \vec{a} \cdot \vec{b} = 0$$

$$|\vec{a}| = |\vec{b}| = 1 \text{ and } |\vec{c}| = 2$$

Also, given $\vec{c}$ is inclined on $\vec{a}$ and $\vec{b}$ with same angle α.

$\therefore \quad \vec{a} \cdot \vec{c} = x|\vec{a}|^2 + y(\vec{a} \cdot \vec{b}) + \vec{a} \cdot (\vec{a} \times \vec{b})$

$$|\vec{a}||\vec{c}| \cos\alpha = x + 0 + 0$$

$$x = 2\cos\alpha$$

Similarly,

$$|\vec{b}||\vec{c}| \cos\alpha = 0 + y + 0$$

$\Rightarrow y = 2\cos\alpha$

$$|\vec{c}|^2 = x^2 + y^2 + |\vec{a} \times \vec{b}|^2$$

$$4 = 8\cos^2\alpha + |a|^2 |b|^2 \sin^2 90^\circ$$

$$4 = 8\cos^2\alpha + 1 \quad \Rightarrow \quad 8\cos^2\alpha = 3$$

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55. For reasons of symmetry, the rectangle will be symmetric with respect to the y axis and thus the two vertices lying on the x axis have coordinates $(-x, 0)$ and $(x, 0)$ and the ones lying on the parabola have coordinates $(-x, 9-x^2)$ and $(x, 9-x^2)$, and because they must lie above the x axis we can consider only $x \in [0, 3)$.

We have to maximize the area:

$A(x) = 2x(9-x^2)$ for $x \in [0, 3)$

$A(x) = 18x - 2x^3$

$\frac{dA}{dx} = 18 - 6x^2 = 6(3-x^2)$

To find the critical points we equate the derivative to zero:

$\frac{dA}{dx} = 0 \Rightarrow x = \sqrt{3}$

and choose the solution in the interval $[0, 3)$

$\frac{d^2A}{dx^2} = -9x$

As for $x = \sqrt{3}$, we have that $\frac{d^2A}{dx^2} < 0$, the critical point is a maximum and the the maximum area is:

$A(\sqrt{3}) = 2\sqrt{3}(9-3) = 12\sqrt{3}$

56. $I = \left(\lim\limits_{n\to\infty} \prod\limits_{k=1}^{n} \left(1+\frac{n}{k}\right)^{k/n}\right)^{1/n}$

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$\left\{\text{Given } f(x) = \left(1+\frac{1}{x}\right)^x \text{ and } f\left(\frac{k}{n}\right) = \left(1+\frac{n}{k}\right)^{k/n}\right\}$

$\log I = \lim\limits_{n\to\infty} \frac{1}{n} \sum\limits_{k=1}^{n} \ln\left(1+\frac{n}{k}\right)^{k/n}$

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$$= \lim_{n\to\infty} \frac{1}{n} \cdot \sum_{k=1}^{n} \frac{k}{n} \ln\left(1+\frac{1}{\frac{k}{n}}\right)$$

$$= \int_0^1 \underbrace{x}_{u} \underbrace{\ln\left(1+\frac{1}{x}\right)dx}_{v}$$

$$= \ln\left(1+\frac{1}{x}\right).\frac{x^2}{2}\Big]_0^1 + \int_0^1 \left(\frac{1}{x}-\frac{1}{x+1}\right).\frac{x^2}{2}dx$$

$$= \frac{1}{2}\ln 2 + \frac{1}{2}\left[x-\ln(x+1)\right]_0^1$$

$$= \frac{1}{2}\ln 2 + \frac{1}{2}\left[(1-\ln 2)-0\right] = \frac{1}{2}$$

$I = \sqrt{e}$

57. $f\left(f\left(f(x)\right)\right) = 1 - f(x)$

$\Rightarrow f(1-x) = 1 - f(x) \Rightarrow f(x) + f(1-x) = 1 \forall x \in [0,1]$

$J = \int_0^1 f(x)dx$

$J = \int_0^1 f(1-x)dx$

$2J = \int_0^1 1.dx \Rightarrow J = \frac{1}{2}$

Also $f''(x) + f''(1-x) = 0$

$\Rightarrow f''\left(\frac{1}{2}\right) = 0$

Given $f''\left(\frac{4}{5}\right) = 0$

$\therefore$ by Rolles theorem,

$f'''(x) = 0$ for some $x \in \left(\frac{1}{2}, \frac{4}{5}\right)$

58.

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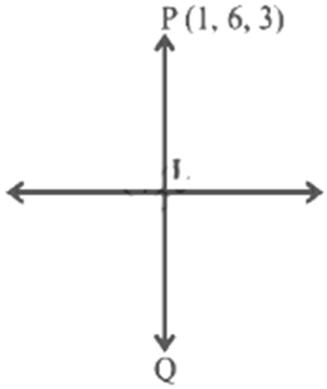
$$\text{Area} = 4\times 2+\frac{2}{3}.4.2$$

$$=8+\frac{16}{3}=\frac{40}{3}$$

59. $dy = d\left(\left(x^2-2x\right)\cos x\right)$

$\therefore y(x)=\left(x^2-2x\right)\cos x+3$

60. Let P(1, 6, 3) be the given point and let L be the foot of perpendicular from P to the given line.



The coordinates of a general point on the given line are

$\frac{x-0}{1}=\frac{y-1}{2}=\frac{z-2}{3}=\lambda$

i.e., $x = \lambda, y = 2\lambda + 1, z = 3\lambda + 2$.

If the coordinates of L are $(\lambda, 2\lambda + 1, 3\lambda + 2)$, then the direction ratios of PL are $\lambda - 1$, $2\lambda - 5, 3\lambda - 1$.

But the direction ratios of given line which is perpendicular to PL are 1, 2, 3.

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Therefore, $(\lambda - 1)1 + (2\lambda - 5)2 + (3\lambda - 1)3 = 0$, which gives $\lambda = 1$.

Hence coordinates of L are (1, 3, 5).

Let $Q(x_1, y_1, z_1)$ be the image of P(1, 6, 3) in the given line.

Then L is the mid-point of PQ.

Therefore,

$\frac{x_1+1}{2} = 1$

$\frac{y_1+6}{2} = 3$

$\frac{z_1+3}{2} = 5$

$\Rightarrow x_1 = 1, y_1 = 0, z_1 = 7$

Hence, the image of (1, 6, 3) in the given line is (1, 0, 7).

61. When $y - x > 8$, the following choices can be made:

Suppose $x = 0$, then y can be any one of $9, 10, 11, \ldots, 20 \rightarrow 12$ ways

Similarly $x = 1 \rightarrow y$ can be $10, 11, \ldots, 20 \rightarrow 11$ ways

$x = 1 \rightarrow y$ can be $11, 12, \ldots, 20 \rightarrow 10$ ways

$\vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots$

$x = 11 \Rightarrow y = 20 \rightarrow 1$ way

$\therefore$ the total number of ways $= 1 + 2 + \ldots\ldots + 12 = 6 \times 13 = 78$

Similarly the number of ways for $x - y > 8$ is 78

Total number of ways for $x - y > 8$ is 78 .

Total number of all possible ways $= 21 \times 21$

Hence the required probability $= \frac{2 \times 78}{21 \times 21} = \frac{52}{147}$

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62. $x^2+3x+2>0$

$\Rightarrow \quad x\in(-\infty,-2)\cup(-1,\infty)$

$x^2+3x+2\le 1$

$\Rightarrow \quad x\in\left[\dfrac{-3-\sqrt{5}}{2},\dfrac{-3+\sqrt{5}}{2}\right]$

$x\in\left[\dfrac{-3-\sqrt{5}}{2},-2\right)\cup\left(-1,\dfrac{-3+\sqrt{5}}{2}\right]$

$0<\sin^{-1}\left(\sqrt{x^2+3x+2}\right)\le\dfrac{\pi}{2}$

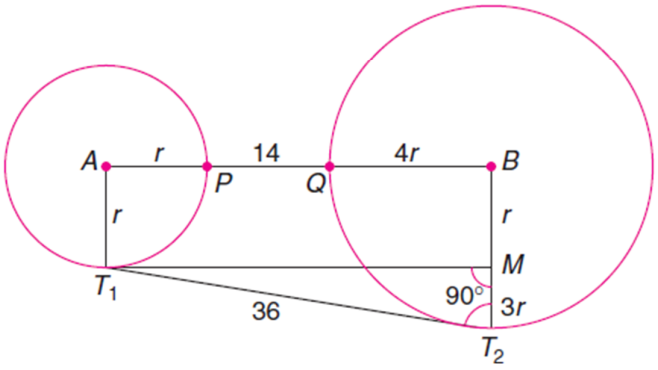
$-\infty<\log\left(\sin^{-1}\left(\sqrt{x^2+3x+2}\right)\right)\le\log\dfrac{\pi}{2}$

$\therefore$ set of all non positive integers.

63. $\Delta=(p^2-4)(q-4),\Delta_1=(p^2-4)(4q-19),\Delta_2=0,\Delta_3=p^2-4.$

64. Points A and B are the centres of the two circles. $PQ = 14$ and $T_1T_2 = 36$. Draw T_1M parallel to AB to BT_2 (see Fig. 3.60). Let r and $4r$ be the radii. From $\Delta\, T_1T_2M$, we have $(T_1M)^2 = (T_1T_2)^2 + (T_2M)^2$. Therefore,

$$(14+5r)^2=36^2+9r^2$$
$$\Rightarrow 16r^2+140r-50\times 22=0$$
$$\Rightarrow 4r^2+35r-275=0$$
$$\Rightarrow 4r^2-20r+55r-275=0$$
$$\Rightarrow 4r(r-5)+55(r-5)=0$$
$$\Rightarrow r=5$$



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65. $\lambda^2+1=(\lambda^2+2)\left(1-\frac{1}{6}\right)$

$\Rightarrow 6(\lambda^2+1)=5\lambda^2+10$

$\Rightarrow \lambda^2=4$

Hence, the ellipse is

$$\frac{x^2}{5}+\frac{y^2}{6}=1$$

and the latus rectum is

$$\frac{2a^2}{b}=\frac{2(5)}{\sqrt{6}}=\frac{10}{\sqrt{6}}$$

66.

Class-interval	f_i	x_i	c.f.	$d_i=\lvert x_i-\text{Med}\rvert$	f_id_i
0–6	4	3	4	11	44
6–12	5	9	9	5	25
12–18	3	15	12	1	3
18–24	6	21	18	7	42
24–30	2	27	20	13	26
	N = 20				$\sum f_id_i$ = 140

Median class $=\left(\frac{N}{2}\right)$th term $=\frac{20}{2}$th term $=10$th term i.e. $12-18$

$\therefore$ Median $= l+\frac{N/2-cf}{f}\times h$

$$=12+\frac{10-9}{3}\times 6=12+\frac{1}{3}\times 6=12+2=14$$

and MD $=\frac{\sum f_id_i}{N}=\frac{140}{20}=7$

Hence, the required MD = 7

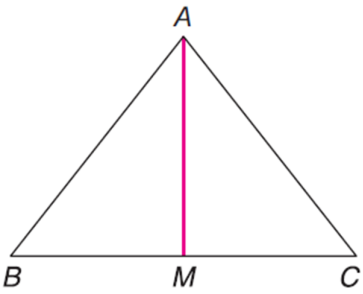
67. $f(\theta)=2+4\cos\theta$

$6\theta\in\left[\frac{\pi}{3},\frac{2\pi}{3}\right]$

$\therefore \max=f\left(\frac{2\pi}{3}\right)=0$

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68.



$\overrightarrow{AB}+\overrightarrow{BC}=\overrightarrow{AC}\Rightarrow\overrightarrow{BC}=\overrightarrow{AC}-\overrightarrow{AB}=2\vec{i}-2\vec{j}$

Suppose AM is the altitude from A to BC

$$\text{Area of } \Delta ABC=\frac{1}{2}|\overrightarrow{BC}\times\overrightarrow{AB}|$$

$$=\frac{1}{2}\sqrt{164}=\sqrt{41}$$

$$\frac{1}{2}(BC)(AM)=\sqrt{41}\Rightarrow\frac{1}{2}\left(2\sqrt{2}\right)(AM)=\sqrt{41}\Rightarrow AM=\sqrt{\frac{41}{2}}$$

69

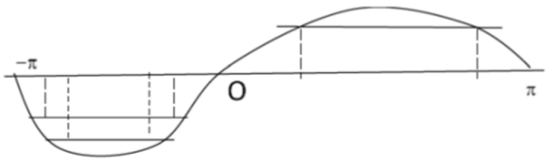
Let $\sin x=t\ \&[-1,1]$

$\therefore 4t^3+2t^2-2t-1=0$

$(2t+1)\left(2t^2-1\right)=0$

$t=\frac{-1}{2}\text{ or }t-\frac{1}{\sqrt{2}}\text{ or }r=\frac{-1}{\sqrt{2}}$

i.e., $\sin x=\frac{-1}{2},\frac{1}{\sqrt{2}}\text{ or }\frac{-1}{\sqrt{2}}$



between $[-\pi,\pi]$ Equation has 6 roots.

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70. $$I=\lim_{n\to\infty}\frac{\int_0^{1/n} f(t)\,dt}{\frac{1}{n^2}}\quad\left(\frac{0}{0}\text{ form}\right)$$

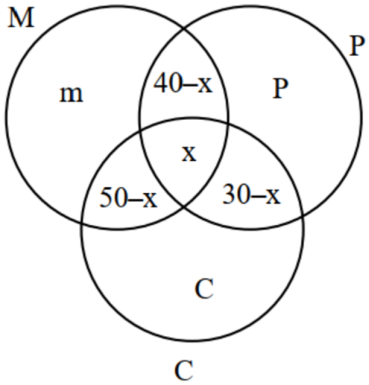
Using : L' Hospital and using Leibnitz rule

$$I=\lim_{n\to\infty}\frac{n^3 f\left(\frac{1}{n}\right)\left(-\frac{1}{n^2}\right)}{-2}=\lim_{n\to\infty}\frac{n}{2}f\left(\frac{1}{n}\right)$$

Put $n=\frac{1}{h}$ hence

$$I=\frac{1}{2}\times\lim_{h\to0}\left(\frac{f(0+h)-f(0)}{h}\right)=\frac{1}{2}f'(0)$$

71. Let M: mathematics; P: Physics; C: Chemistry



$125 \le m + 90 - x \le 130$...(i)

$85 \le P + 70 - x \le 95$...(ii)

$75 \le C + 80 - x \le 90$....(iii)

Also,

$m + P + C + 40 - x + 30 - x + 50 - x + x = 210$

$\Rightarrow m + P + C + 120 - 2x = 210$

$\Rightarrow m + P + C - 2x = 90$

$\Rightarrow 15 \le x \le 45$ & $30 - x \ge 0$

$\Rightarrow 15 \le x \le 30$

So, m = 15 and n = 30

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72. $[x]=0, \forall x \in [0,1)$

For $x \in [1,2), [x]=1$

$\Rightarrow \frac{[x]}{1+x^2}=\frac{1}{1+x^2}<1, \forall x \in [1,2) \Rightarrow \left[\frac{[x]}{1+x^2}\right]=0$

For $x \in [-1,0), [x]=-1 \Rightarrow \frac{[x]}{1+x^2}=-\frac{1}{1+x^2}$

Clearly, $2 \geq 1+x^2>1, \forall x \in [-1,0)$

$\frac{1}{2} \leq \frac{1}{1+x^2}<1 \Rightarrow -\frac{1}{2} \geq -\frac{1}{1+x^2}>-1$

$\Rightarrow \left[\frac{[x]}{1+x^2}\right]=-1 \forall x \in [-1,0)$

$\therefore$ Given integral $= \int_{-1}^{2} 50dx + \int_{-1}^{0} -1dx + \int_{0}^{2} 0dx = 150-1=149$

73. $\left(x^2-5\right)\frac{dy}{dx}-2xy=-2x\left(x^2-5\right)^2$

$\frac{dy}{dx}+\left(\frac{-2x}{x^2-5}\right)y=-2x\left(x^2-5\right)$

I.F. $= \frac{1}{\left|x^2-5\right|}$

Solution of D.E. is $y.\frac{1}{\left|x^2-5\right|}=\int -2x.\frac{x^2-5}{\left|x^2-5\right|}dx \Rightarrow \frac{y}{\left|x^2-5\right|}=\frac{x^2-5}{\left|x^2-5\right|}\left(-x^2\right)+C$

$\because y(2)=7 \Rightarrow C=3$

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74. $A(G-H)x^2 + G(H-A)x + H(A-G) = 0$

Let α and β be roots of the given equation

$\alpha < \beta$

$\therefore \alpha + \beta = \dfrac{-G(H-A)}{A(G-H)} \Rightarrow \alpha\beta = \dfrac{H(A-G)}{A(G-H)}$

$\beta = 1$ is satisfied the equation,

i.e. $AG - AH + GA - GA + HA - HG = 0$

$\therefore$ One of root is 1

$$\alpha = \frac{H(A-G)}{A(G-H)}$$

$$\Rightarrow \quad \alpha = \frac{AH - GH}{AG - AH}$$

$$\alpha = \frac{G^2 - GH}{AG - AH} \qquad [\because AH = G^2]$$

$$\alpha = \frac{G(G-H)}{A(G-H)} \Rightarrow \alpha = \frac{G}{A} \qquad [\because A > G]$$

$\alpha < 1$

Hence, $0 < \alpha < 1$

75. $a = \sum_{n=1}^{10}\left(n^2 + 3n + 1\right)^2 = \sum_{n=1}^{10}\left(n^4 + 6n^3 + 11n^2 + 6n + 1\right)$

$\Rightarrow a - b = \sum_{n=1}^{10}\left(6n^3 + 11n^2 + 6n + 1\right) = 22725$

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